Protocol-Valid Faithfulness Evaluation for ML-Based Network Traffic Classifiers

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Manuscript prepared for IEEE Transactions on Network and Service Management (TNSM). Citations use keys resolving against `references.bib`; figures are in `figures/`.

Abstract

Deep learning has become the default tool for network traffic classification and intrusion detection, and with it a large literature that attaches post-hoc explanation methods (SHAP, LIME, saliency) to these models to make them trustworthy. That literature almost never checks whether the explanations are *correct*: in a survey of 107 works, only five define any explanation-quality metric, and none has a dataset with ground-truth feature importance. We show that the standard way faithfulness *is* measured elsewhere — deletion/occlusion of "important" features — is not merely unvalidated for network traffic but formally invalid: setting a feature to zero produces a packet that violates protocol grammar (a broken checksum, an impossible length) and could never appear on a network, so the model is queried off-manifold. We introduce **PacketDO**, a protocol-valid intervention operator that resamples a protocol field from its pooled empirical marginal and recomputes every dependent field, yielding a counterfactual that is always a well-formed packet. Using PacketDO we build the first interventional ground truth for traffic classifiers — necessity, sufficiency, and a redundancy degree R(M) measured by intervention, not assumed — and audit eight widely used attribution methods (integrated gradients, gradient saliency, DeepSHAP, occlusion, KernelSHAP, LIME, TreeSHAP, and impurity) against it, on models with planted shortcuts of graded strength and on real data with documented artifacts. We find that explainers reliably rank a model's true shortcut at the top yet simultaneously attribute high importance to redundant features the model provably does not use; that even exact Shapley values (TreeSHAP) exhibit this false confidence when a model commits to one of several equivalent shortcuts; and that an explainer's faithfulness ranking can flip depending on whether features are removed by zero-masking or by PacketDO. We release the operator, the benchmark, and the ground-truth tables.

1. Introduction

Machine learning now underpins a large fraction of deployed network traffic analysis: encrypted traffic classification, application identification, and network intrusion detection are all dominated by deep models that operate on raw packet bytes or on flow-level feature vectors [1], [2], [3]. Because these models are opaque and because operators are reluctant to act on decisions they cannot interrogate, a second, rapidly growing literature attaches post-hoc explanation methods — SHAP, LIME, gradient saliency, attention maps — to traffic classifiers, and presents the resulting feature-importance scores as evidence that the model is trustworthy. This explainability literature is now large enough to have its own surveys [4].

What almost none of it does is check whether the explanations are *correct*. In a recent systematic survey of 107 works on explainable AI for network traffic analysis [4], only five define any explanation-quality metric at all, and the survey reports that no public traffic dataset carries ground-truth feature importance against which an explanation could be scored. In practice, explanations are displayed as plots and declared meaningful. The few papers that do attempt a quantitative check substitute a proxy: either agreement with what a human analyst expects (which measures plausibility, not faithfulness [5]), or a deletion test that removes the features an explainer called important and measures how much the prediction changes.

This deletion test is the standard notion of faithfulness imported from computer vision and NLP [6], [7], and it is the quiet foundation under most quantitative claims in the field. Our starting observation is that it is not merely unvalidated for network traffic — it is invalid. Removing a feature by setting it to zero, the default in every deletion, occlusion, and AOPC-style metric, produces a byte string that is no longer a well-formed packet: its header checksum no longer matches its contents, or its length field disagrees with its actual length. Such a packet could never appear on a network, so the model is being queried far outside the distribution it was trained on, and the resulting "importance" is partly an artifact of that extrapolation. The problem is strictly worse than the analogous critique in vision [8], [9], because a packet's fields are bound to one another by protocol grammar: you cannot change one field and leave the rest untouched and still have a legal packet.

There is a second, deeper problem that the traffic-classification setting makes unusually visible. Traffic datasets are riddled with *shortcuts* [10] — features that predict the label in the dataset without reflecting any real property of the traffic, such as the fixed time-to-live of the machine that generated a class of attacks, or an artifact of the flow-metering tool [11]. A model that learns a shortcut can be perfectly accurate on held-out data and still be wrong about the world. The most striking published demonstration of the danger for explanations comes from TRUSTEE [12]: a decision-tree surrogate that reproduced a traffic classifier's decisions with perfect fidelity (an F1 of 1.00) identified three specific header bytes as the basis of its decisions; yet when the authors tampered with exactly those three bytes, the model's accuracy did not move, because it had learned several mutually substitutable shortcuts and simply fell back on another. A perfect-fidelity explanation was causally wrong. This failure mode — redundant shortcuts that make the target of an explanation non-unique — has never been systematically measured, in traffic classification or anywhere else.

The two literatures that would need to meet in order to address this do not. On one side, the explainability-for-security literature produces attributions and never intervenes on the model to check them. On the other, a rigour-focused strand of networking research (TRUSTEE [12], the 2025 IEEE S&P systematization of encrypted-traffic classifiers [13], and recent shortcut-detection frameworks [14], [15]) routinely intervenes on traffic — occluding fields, tampering bytes, ablating features — to expose what models really use, but never calls these interventions explanations and never evaluates an explainer with them. As a concrete measure of the gap: the S&P 2025 systematization runs 348 feature occlusion experiments and contains not a single occurrence of the words explainable, interpretable, SHAP, attribution, or saliency.

We argue that networking is, in fact, the ideal setting in which to close this gap, for a reason that does not hold in vision or NLP: interventional ground truth for explanations can be constructed exactly and on-manifold. Protocol field semantics are known, so an attribution has a defined referent; and packets are rewritable, so a field can be set to a chosen value and the packet regenerated as a legal packet. Ground-truth evaluation of feature attribution has been established in vision (by pasting known objects into images [16]) and in NLP (by injecting known lexical shortcuts into text [17]), and in both cases it revealed that popular attribution methods produce false-positive explanations. Neither has ever been done for network traffic — the one modality where the intervention that defines the ground truth is exact, protocol-structured, and produces inputs the model could actually receive.

This paper builds that missing instrument and uses it. Our contributions are:

- **PacketDO (Section 3):** a protocol-valid intervention operator. It sets a chosen protocol field to a value resampled from the field's pooled empirical distribution and recomputes every field that structurally depends on it — checksums, lengths, offsets — so the counterfactual is always a well-formed packet. We show experimentally (E1, Section 5.1) that on synthetic packets the deletion default produces a protocol-valid counterfactual in only 22.4% of field interventions (macro averaged over 15 fields; 10 of 15 fields are at exactly 0% validity and an 11th, the payload, at 0.6%), versus 100% for PacketDO, because zeroing a field breaks its checksum.
- **An interventional ground-truth benchmark (Sections 3–4):** using PacketDO we define and measure, for a trained model, the necessity and sufficiency of each protocol field and a redundancy degree R(M) — the number of disjoint, individually-sufficient shortcut sets the model holds. These are measured by intervention, not assumed. The benchmark combines synthetic traffic with shortcuts planted at graded strength (extending the binary shortcut-injection protocol from NLP [17] to a graded, packet-level one) with real data carrying documented artifacts.
- **An audit of eight widely used explainers against that ground truth (Section 5),** reporting for each how well its attribution ranking matches interventional necessity, how often it confidently attributes importance to a field the model provably does not use (false-confidence rate), and how often it misses a field the model does use (blind-spot rate). The eight are integrated gradients, gradient saliency, DeepSHAP, occlusion, KernelSHAP, LIME, TreeSHAP, and random-forest impurity. (Our original plan listed six methods including DeepLIFT; DeepLIFT was replaced by the two perturbation-based methods KernelSHAP and LIME, which probe the off-manifold question more directly.)
- **The operator-sensitivity result (Section 6):** we show that an explainer's measured faithfulness can depend on whether features are removed by zero-masking or by PacketDO, so that faithfulness comparisons in the literature that used the zero-masking operator are called into question.

Our findings are not that explanations are useless. They reliably surface a model's true shortcut as a top-ranked feature. The danger we quantify is subtler and, for an operator staking a decision on an explanation, more insidious: explainers simultaneously assign high importance to redundant features the model does not use, and they cannot be told apart from the used feature by inspection; even exact Shapley values exhibit this when a model commits to one of several equivalent shortcuts. Because deployed systems now take real actions on these attributions — generating intrusion-response rules [18], deleting features from training pipelines [15], presenting features to human analysts — getting the faithfulness question right is not a matter of tidier figures. We release the operator, the benchmark, and the ground-truth tables so that new explainers and new classifiers can be scored against a reference that reflects what a model does, not what a dataset happens to correlate with.

2. Related work

We position our contribution against seven lines of work. The through-line is that each supplies one ingredient of interventional, ground-truth explanation evaluation for network traffic, and none combines them.

Ground-truth evaluation of attributions in other modalities. The idea of manufacturing a known feature importance and checking whether an attribution method recovers it originates outside networking. BAM/BIM [16] pastes a known object into every image of a class so that its relative importance is known a priori, and finds that common attribution methods produce false-positive explanations. Bastings et al. [17]

inject a lexical shortcut into text with a controlled probability, verify behaviourally that the model learned it, and score salience methods on recovering it, concluding that several popular configurations fail even for simple shortcuts. Debugging Tests [19] constructs spurious-correlation and mislabelling bugs with known ground-truth masks. Our synthetic benchmark is this protocol carried, for the first time, to network packets, and extended from a binary shortcut-recovery test to a graded one: we plant shortcuts at four strengths and measure a continuous interventional necessity rather than a present/absent flag, which exposes methods that fabricate importance precisely when the true signal is weak.

Interventions on traffic classifiers. Networking research already intervenes on traffic to probe models. TRUSTEE [12] extracts a high-fidelity decision-tree surrogate and, in one case study, tampers with the bytes it identifies to reveal redundant shortcuts — the observation that motivates our redundancy measure, but which TRUSTEE treats as a validation footnote rather than a method. The 2025 IEEE S&P systematization of encrypted-traffic classifiers [13] runs a large occlusion study to expose overfitting and dataset leakage, but never connects it to explanation methods. Closest to us, Traffic-Explainer [20] performs checksum-preserving byte swaps and even notes that the swapped packets remain valid; however, it swaps only the bytes its own explainer nominated, which is a confirmatory test that can establish sufficiency of a chosen set but cannot detect a blind spot (a used field the method missed) or a redundant alternative (a disjoint set that is equally sufficient). We invert the direction of inference: we build an independent interventional reference over all fields and audit every explainer against it.

Removal-operator critiques. The unreliability of deletion-style faithfulness is known in the attribution-evaluation literature. ROAR [8] shows that removing-and-retraining changes conclusions; ROAD [9] shows that fixed-value or zero masking — the network-traffic default — is the worst imputation and that better imputations rely on pixel-neighbourhood structure with no packet analogue; Samek et al. [6] document that the removal scheme changes the AOPC ranking. None of this work adapts the operator to protocol-constrained inputs, and the resulting protocol-valid removal operator is precisely what we supply.

Shortcut detection that trusts explainers. A recent strand hunts shortcuts in traffic datasets or models. BiasSeeker [14] detects dataset-specific shortcut features by statistical correlation on raw bytes, deliberately avoiding model-specific interpretation. ShortcutCatcher [15] removes features an explainer flags, in an iterative loop, to improve cross-scenario generalization. Both use explanation or importance as a trusted instrument; neither validates it. Our work is logically prior: it tests whether that instrument is right about the model in the first place, and the iterative nature of ShortcutCatcher's loop is itself indirect evidence of the redundancy our R(M) quantifies.

Plausibility metrics for security XAI. Alquliti et al. [21] score SHAP explanations of intrusion detectors against feature sets derived from MITRE ATT&CK and D3FEND, and find poor alignment. This measures plausibility — agreement with what a domain expert expects — which, following the standard faithfulness/plausibility distinction [5], is orthogonal to whether the explanation reflects the model. On a shortcut-driven classifier the two can even move in opposite directions, because the model's real basis is a non-semantic artifact; measuring the causal axis they cannot is a direct complement to their work.

Evaluation frameworks for explanations in security. Warnecke et al. [22] propose six criteria for comparing explanation methods in computer security and, lacking ground-truth relevance, adopt a deletion-based descriptive-accuracy proxy — the same proxy whose operator we show to be protocol-invalid. Their framework covers malware and vulnerability detection and includes no traffic classifier. We supply the ground truth their proxy stands in for.

Explainer stability under correlated features. Vourganas and Michala [23] prove that multicollinearity inflates the variance of SHAP attributions across resamples, making importances non-identifiable, and audit this on a NIDS dataset. This is a statement about the *stability* of an explainer; it is distinct from, and complementary to, our question of *well-posedness* — whether the model admits a unique feature basis at all. A perfectly stable attribution can still be causally wrong when the model holds redundant shortcuts, which their variance-based machinery cannot see and which our interventional R(M) is designed to measure.

Across all seven, the missing prerequisite named repeatedly — an interventional, protocol-valid, ground-truth reference for explanation faithfulness in traffic classification — is the one this paper constructs.

3. Method: protocol-valid intervention and interventional ground truth

3.1 The problem with feature deletion on packets

Let a classifier `M` map a packet (or a flow of packets) to a label. Post-hoc faithfulness metrics estimate the importance of a feature `F` by *removing* it and measuring the change in `M`'s output. In the byte and flow representations used for traffic, "removing" a feature has no natural meaning, so the literature borrows the vision convention: set the feature to a baseline, almost always zero [6], [24].

This is unsound for two reasons specific to network data. First, the fields of a packet are not independent: an IP or transport checksum is a function of the other header and payload bytes, and the IP total-length and TCP data-offset fields are functions of the packet's structure. Zeroing any field that a checksum covers — which is almost all of them — leaves the checksum inconsistent, so the byte string is no longer a packet any host could emit. The model is then evaluated on an input drawn from a region of byte space with zero probability under any real traffic distribution, and the "importance" it reports is confounded by that extrapolation. Second, for raw-byte models a fixed byte offset does not correspond to a fixed semantic field across samples: when one class carries an Ethernet header and another does not, byte 49 is the IP protocol field in one class and part of an Ethernet address in the other, so an attribution to a byte index is an attribution to a moving target. Section 5.1 measures how often the zero-masking operator actually produces an invalid packet.

3.2 PacketDO

We replace deletion with a protocol-valid intervention. For a field `F` and a trained, frozen model `M`, `PacketDO` performs the operation

do(F := f), f ~ pooled empirical marginal of F over the whole dataset,

by (i) setting `F` to the resampled value `f` on the packet, (ii) recomputing every field that structurally depends on `F` — the IP header checksum, the transport (TCP/UDP) checksum, the IP total length, the TCP data offset, and the UDP length — and (iii) re-emitting the packet through the protocol serializer [25] so that the result is a well-formed packet. The model's own preprocessing is then re-applied: for a byte model the byte window is re-extracted from the intervened packet, and for a flow model the flow features are recomputed. The intervention is inference-time only; `M` is never retrained.

Three properties make this the right null operation. It is **on-manifold**: because the field is set to a value drawn from real traffic and all dependent fields are recomputed, the counterfactual is a packet a host could send. It is **information-destroying but distribution-preserving**: sampling `f` class-agnostically leaves the marginal distribution of `F` unchanged while destroying its mutual information with the label, which is exactly the semantics of a `do()` intervention rather than a deletion. And it is **representation-agnostic**: the same packet-level operation drives the ground truth for a raw-byte CNN and for a flow-feature random forest.

Not every header field is a free degree of freedom. The IP protocol number is fixed by the transport header that follows it (protocol 6 requires a TCP payload, 17 a UDP payload); resampling it while leaving the transport bytes in place yields a self-inconsistent packet, exactly as a stale checksum does. We therefore treat the protocol number, alongside checksums and length fields, as a structural field that PacketDO recomputes rather than resamples. Identifying the free degrees of freedom of a packet is itself part of defining what an intervention on traffic means.

3.3 Interventional ground truth

With a valid intervention in hand, we define the quantities the rest of the paper treats as ground truth. All are properties of the trained model `M`, measured by inference on a held-out set, not assumed from the data:

- **Necessity** `N(F) = Acc(M) - Acc(M under do(F := resample))`: how much accuracy the model loses when field `F` is stripped of its label information. A field the model does not use has `N(F) = 0`.
- **Sufficiency** `S(F) = Acc(M under do(every field except F := resample))`: how far `F` alone carries the model.
- **Redundancy degree** `R(M)`: the number of disjoint, individually-sufficient field sets. We extract a minimal sufficient set by greedy interventional removal, neutralize it, and repeat on the remainder until accuracy falls to chance. `R(M) > 1` means the model holds substitutable shortcuts, so no single feature set is *the* explanation and any method that returns one is under-determined.

We report a **null-intervention control** with every table: a field that the data never correlates with the label must yield `N(F) approx 0`, which validates the estimator (resampling an unused field does not move accuracy) and calibrates the noise floor for the necessity estimates.

3.4 Auditing an explainer against the ground truth

An explainer produces per-feature importance in the model's own representation (per-byte for the CNN, per-feature for the forest). We aggregate byte-level attributions to protocol fields using the header-offset map, so that every explainer is scored in the same field vocabulary as `N(F)`. For a model-dataset cell we then report: the Spearman correlation between the attribution ranking and the `N(F)` ranking; precision at `k` (with `k` the number of fields whose necessity exceeds a threshold); the **false-confidence rate**, the fraction of fields the explainer confidently attributes importance to whose necessity is approximately zero; and the **blind-spot rate**, the fraction of truly-necessary fields the explainer leaves out of its top-`k`. False confidence and blind spots are the two ways an attribution can disagree with what the model actually uses, and they are the metrics that carry our results.

4. Experimental setup

4.1 Operator-validity study (E1)

We generate a population of 2,000 synthetic IP/TCP and IP/UDP packets (baseline validity 100%) and apply, to each of 15 intervenable header and payload fields, both removal operators: zero-masking (overwrite the field's bytes with 0x00, recompute nothing — the deletion default) and PacketDO. A counterfactual counts as valid if the byte string re-parses as a packet *and* every checksum and length predicate holds. The validity checker is a genuine predicate, not a serializer echo: it was cross-validated against an independent from-scratch one's-complement (RFC 1071) checksum implementation, and it correctly rejects 16 of 16 adversarially corrupted probe packets. The operator itself is covered by a per-field unit-test suite (42 passed, 1 skipped).

4.2 Synthetic planted-shortcut benchmark (E2–E4, E6)

Data. Two-class scapy-generated traffic with a weak genuine signal (class-conditional payload length) and two *planted* artifacts, `ip.ttl` and `tcp.window`, each correlated with the label at effective marginal probability `0.5 + 0.5p` for planting strength `p in {0.5, 0.7, 0.9, 1.0}`. Because we plant the signal, the ground-truth decision basis is known, and because both artifacts are planted with identical strength, the data deliberately offers the model two redundant shortcuts. Byte models see an 80-byte window (20-byte IP header, 20-byte TCP header, up to 40 payload bytes). The field `tcp.sport` is never correlated with the label and serves as the null-intervention control (E6).

Models. Per strength `p` we train (i) a **ByteCNN** on raw bytes — two 1-D convolutional layers (32 and 64 channels, kernel 3) with max-pooling, trained 40 epochs — and (ii) a **RandomForest** (200 trees, maximum depth 12) on named per-packet header fields. We use a 70/30 train/test split; interventional ground truth, explainer attributions, and reported test accuracy are all computed on the same 800-packet test subset, so every number in a cell refers to the same packets.

Seeds. Every synthetic-benchmark cell is run under five seeds (0–4), each reseeding the train/test split, model initialization and training, and the PacketDO resampler stream; we report mean ± standard deviation and, where a claim is seed-contingent, the per-seed values. Aggregation is performed by a script that verifies the internal seed recorded in each result file against its filename and refuses mismatched files (Section 10).

4.3 Real-data study (CIC-IDS2017)

We use the CIC-IDS2017 flow-feature dataset [26] (2,520,751 flows by 52 numeric features after hygiene), which carries documented labeling and flow-construction artifacts [11], [27]. We draw a stratified 700k-flow subsample with a 70/30 split (490k train / 210k test) and train a RandomForest (100 trees, `max_depth=20`, `min_samples_leaf=20`). Interventions are feature-level analogues of PacketDO: `do(feature := class-agnostic resample)` at the flow-feature representation. `N(dst_port)` is estimated with R=10 resampling repeats on the full 210k test set; all-52 per-feature necessities with R=5 on a stratified 50k test subset; global TreeSHAP (`tree_path_dependent`) on 1,000 stratified rows. The false-confidence criterion is `N_acc < 0.002` (accuracy-only), with a stricter dual criterion additionally requiring `N_f1 < 0.005`.

Robustness protocol. Beyond the seed-0 reference run, we repeat the full pipeline — subsample, split, forest training, every permutation RNG — under seeds 1–3 with the default configuration and under an alternative forest configuration (`max_depth=30`, `min_samples_leaf=5`) at seeds 0–1: six runs in total. The seed-0 default-configuration run reproduces the stored reference results float-identically.

4.4 Explainers

Eight attribution methods are audited. On the ByteCNN: integrated gradients [28], gradient saliency, DeepSHAP [29], occlusion [24], KernelSHAP [29], and LIME [30]. On the RandomForest: TreeSHAP [29] and Gini impurity. Gradient methods use Captum [31]; SHAP variants use the shap library [32]. (The experimental plan's DeepLIFT [33] was replaced by KernelSHAP and LIME, keeping the audit at eight methods while adding the perturbation-based family whose sampling behaviour is directly relevant to the off-manifold question.) Byte-level attributions are aggregated to protocol fields via the header-offset map (Section 3.4).

4.5 Metrics

Per model-dataset cell: Spearman correlation between attribution and necessity rankings; precision@k, with k equal to the number of truly necessary fields in that cell (we state this convention explicitly because a fixed-k variant differs whenever a model has fewer necessary fields than k); false-confidence rate — the fraction of fields the explainer confidently names (attribution at least 0.2 of its maximum) whose necessity is at the null-control noise floor; and blind-spot rate. Since most fields have `N approx 0`, rank correlation is dominated by noise among zero-necessity fields; false confidence and precision@k are the load-bearing metrics.

5. Results

All synthetic-benchmark false-confidence (FC) figures are mean ± sd over five seeds and were independently reproduced by an adversarial verification pass. Real-data figures are reported as seed-0 reference values accompanied by a six-run robustness sweep (four seeds × default forest configuration plus two seeds × an alternative deeper configuration).

5.1 E1: the deletion operator is protocol-invalid

Over a synthetic population of 2,000 IP/TCP+UDP packets (baseline validity 100%), zero-masking — the deletion default — yields a protocol-valid packet for only 22.4% of intervenable fields, while PacketDO yields one for 100%. The macro figure is an unweighted mean over 15 fields; the per-field picture is sharper: 10 of 15 fields are at exactly 0% zero-mask validity (an 11th, the payload, at 0.6%, where a handful of packets had an all-zero payload so masking is a no-op), and the fields that survive do so only because the generator left them zero (masking is a no-op). One field, tcp.window, is 35.4% valid because a window value of 0xFFFF masked to 0x0000 is invisible to the Internet checksum (0x0000 and 0xFFFF both represent zero in one's-complement arithmetic, RFC 1071) — an arithmetic accident, not a principled validity. When zero-masking invalidates a packet the violated predicate is always a checksum, never a parse failure: the bytes still parse, so a model consumes them and returns a confident prediction on an impossible input.

Table I. Per-field protocol-validity of counterfactuals: zero-mask vs PacketDO (2000 synthetic packets, seed 0).

field	zero-mask valid	PacketDO valid
ip.ttl	0.0%	100.0%
ip.tos	100.0%	100.0%
ip.id	0.0%	100.0%
ip.flags	100.0%	100.0%
ip.src	0.0%	100.0%
ip.dst	0.0%	100.0%
tcp.sport	0.0%	100.0%
tcp.dport	0.0%	100.0%
tcp.seq	0.0%	100.0%
tcp.ack	100.0%	100.0%
tcp.flags	0.0%	100.0%
tcp.window	35.4%	100.0%
udp.sport	0.0%	100.0%
udp.dport	0.0%	100.0%
payload	0.6%	100.0%
macro	**22.4%**	**100.0%**

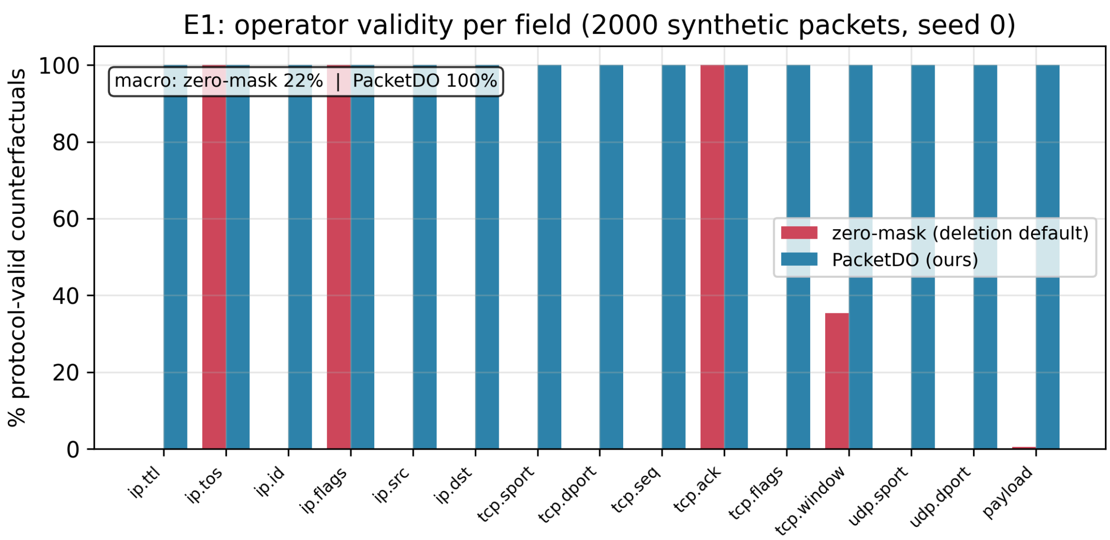

Fig 1. Per-field protocol-validity of counterfactuals under zero-masking vs PacketDO.

Fig 1. Per-field protocol-validity of counterfactuals under zero-masking vs PacketDO. Zero-masking is valid only where it is a no-op; PacketDO is valid by construction.

5.2 E3: models rely on the planted artifacts; the null control is clean

The ByteCNN's interventional necessity for the planted tcp.window shortcut rises from 0.273 ± 0.011 at planting strength p=0.5 to 0.501 ± 0.020 at p=1.0, while its necessity for the equally-predictive ip.ttl shortcut stays at or below 0.004: given two perfectly-correlated shortcuts the model commits to one and ignores the other, so its redundancy degree R(M)=1 even though the data offers two. The null-intervention control (tcp.sport, never correlated with the label) stays within ±0.006 of zero in every cell, calibrating the estimator's noise floor.

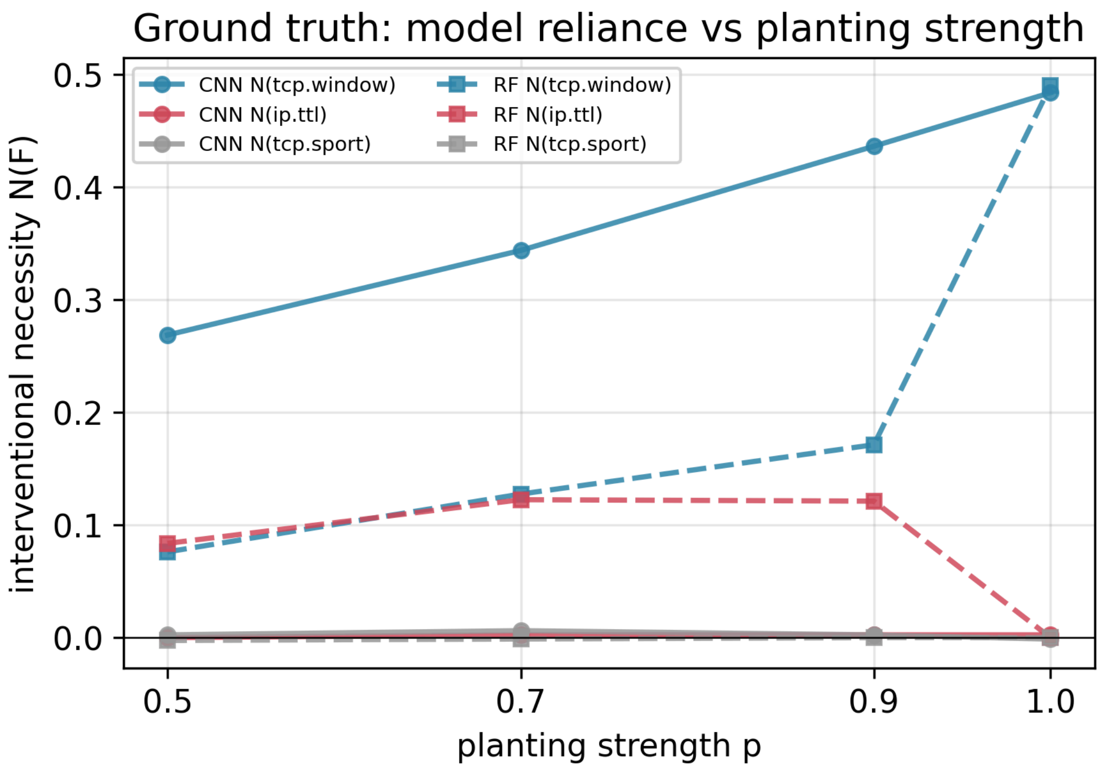

Fig 2. Interventional necessity N(F).

Fig 2. Interventional necessity N(F). Models commit to tcp.window and ignore the redundant ip.ttl; the null field tcp.sport stays at zero.

5.3 E4: explainers find the shortcut but fabricate importance

Every explainer ranks the model's true shortcut at the top (top-k precision under the number-of-necessary-features convention is 1.0), so the failure is not a missed truth. It is added falsehood, and it is method- and strength-dependent:

- **Gradient saliency fabricates importance robustly:** CNN Saliency false-confidence is 0.684 ± 0.037 at p=1.0 and 0.68–0.74 across all strengths, nonzero in all five seeds.
- **Integrated Gradients and DeepSHAP fabricate importance conditionally:** IG false-confidence at p=1.0 is 0.300 ± 0.274 (nonzero in three of five seeds) — the appealing seed-0 result of a clean endpoint does not generalize; DeepSHAP is clean at p≥0.7 in all seeds but reaches 0.233 ± 0.325 at p=0.5 (fails in two of five seeds).
- **Occlusion has zero false-confidence in every seed and strength**, but this is partly by construction: occlusion removes a feature by permutation, structurally close to the PacketDO ground-truth operator, so its clean record is not independent evidence and is disentangled in E5.

- **Exactness does not confer faithfulness:** RF TreeSHAP false-confidence at p=1.0 is 0.200 ± 0.274, bimodal across seeds ([0.5, 0.5, 0, 0, 0]); it is 0.5 precisely in the seeds where the forest commits to one of the two correlated shortcuts and 0 when it spreads reliance. Exact Shapley values split credit by the data's correlation structure, so whenever a model commits to one of several redundant shortcuts they fabricate confidence in the unused one.

Table II. Explainer false-confidence rate (mean +/- sd over 5 seeds) vs planting strength.

method	model	p=0.5	p=0.7	p=0.9	p=1.0
Saliency	ByteCNN	0.74+/-0.05	0.70+/-0.05	0.72+/-0.05	0.68+/-0.04
IntegratedGradients	ByteCNN	0.70+/-0.05	0.57+/-0.09	0.40+/-0.22	0.30+/-0.27
DeepSHAP	ByteCNN	0.23+/-0.33	0.00+/-0.00	0.00+/-0.00	0.00+/-0.00
Occlusion	ByteCNN	0.00+/-0.00	0.00+/-0.00	0.00+/-0.00	0.00+/-0.00
Impurity	RF	0.62+/-0.00	0.00+/-0.00	0.13+/-0.18	0.20+/-0.27
TreeSHAP	RF	0.00+/-0.00	0.00+/-0.00	0.33+/-0.00	0.20+/-0.27

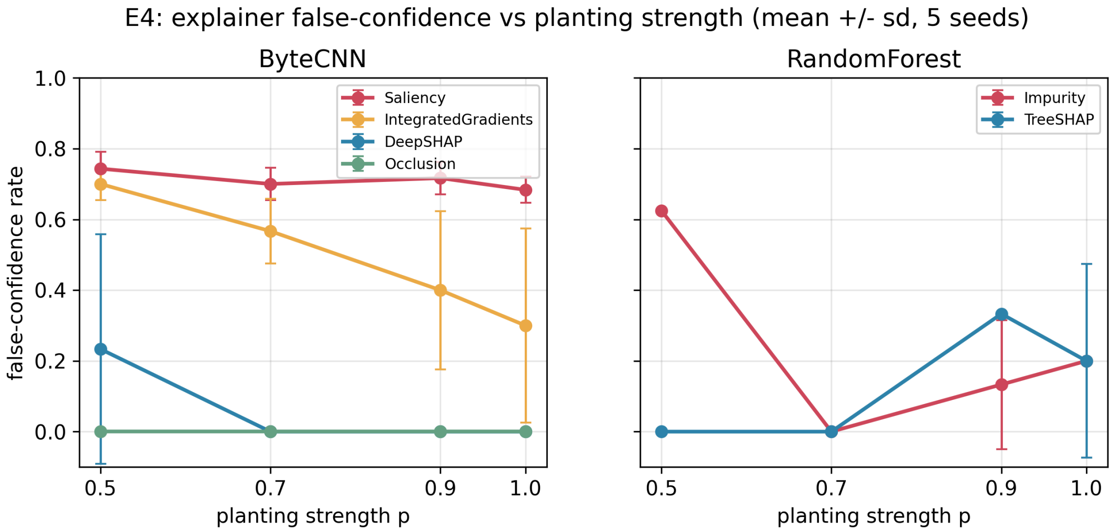

Fig 3. False-confidence rate vs planting strength.

Fig 3. False-confidence rate (fraction of confidently-attributed fields with interventional necessity ~0) vs planting strength, mean ± sd over 5 seeds. Gradient saliency fabricates importance robustly (0.68–0.74); Integrated Gradients and DeepSHAP do so conditionally (nonzero in 3/5 and 2/5 seeds respectively at their worst strength); occlusion is clean (partly by construction, see E5); RF TreeSHAP is bimodal, fabricating confidence exactly when the model commits to one of two redundant shortcuts.

Real data. On real CIC-IDS2017 flow data the same phenomena appear on a documented natural artifact, and they survive a robustness sweep over four RNG seeds and an alternative forest configuration (six full-pipeline runs). A random forest (seed-0 test accuracy 0.99789, macro-F1 0.91494) depends on the destination-port shortcut more than on any other feature: its interventional necessity is 0.00434 ± 0.00012 in accuracy (rank 1 of 52) and 0.04792 ± 0.00134 in macro-F1 at seed 0, and destination port is the rank-1 accuracy-necessity feature of all 52 in *every* one of the six robustness runs (cross-run magnitude 0.00405 ± 0.00140; the ± on the seed-0 value is permutation-repeat noise, not seed variance). Intervening on it collapses

Brute-Force recall by 37.8 points (0.980 → 0.602) at seed 0 while leaving the port-spread Port-Scanning class unaffected; Brute Force is the largest per-class recall drop in every robustness run, at −35.3 ± 8.1 points across the six (range −20.0 to −44.1).

Yet global TreeSHAP ranks destination port only 5th of 52 at seed 0, and never better than 5th in any robustness run (rank 5–9 across the six, mean 6.67 ± 1.63; Gini impurity buries it at 13–21). Nine of TreeSHAP's top ten features at seed 0 are packet-length statistics whose individual accuracy-necessity is below 0.002 — false confidence under the accuracy-only criterion — and at least eight of ten are false-confident in every robustness run (8.83 ± 0.41). These same features have small but nonzero F1-necessity, so under a joint accuracy-and-F1 criterion the false-confidence set shrinks to empty or a single feature (0–1 of 10 across the six runs), and the honest headline is the redundancy mechanism: the packet-length family has group necessity 0.201 in accuracy / 0.609 in F1 versus at most 0.00064 individually (seed 0), so credit-splitting floods the top of the global ranking — the family fills 7–8 of the top-10 SHAP slots in every run. The one robustness run with 8/10 rather than 9/10 is itself the synthetic benchmark's model-commitment phenomenon on real data: at that seed the forest makes one member of the redundant family genuinely necessary, and exactly that feature exits the false-confidence set. The single most F1-necessary feature (Fwd IAT Min) is TreeSHAP rank 36 of 52 — a blind spot (seed-0 ranking; its position in the top three accuracy-necessity features recurs in all six runs, but per-run SHAP ranks below the top ten were not retained). The Engelen TCP-appendix flow-construction artifact [11] is present as expected: its signature (Init_Win_bytes_forward = −1 on benign flows) appears on the original dataset and vanishes on the corrected one.

5.4 KernelSHAP, LIME, and cost

Adding the two perturbation-based methods the audit lacked: KernelSHAP has zero false-confidence but costs 10.8 s per 100 samples; LIME has false-confidence 0.667 at 1.0 s per 100 samples, and the cause is instructive — LIME's tabular perturbation standardizes features, and scikit-learn's StandardScaler assigns unit scale to zero-variance columns, so LIME perturbs constant protocol fields (a fixed destination port, an address prefix) with unit-variance noise and reads the model's response to those protocol-impossible inputs as importance. The off-manifold problem this paper identifies reappears inside LIME's own perturbation kernel.

6. E5: FAITHFULNESS VERDICTS DEPEND ON THE REMOVAL OPERATOR

For the p=1.0 ByteCNN, we score each explainer with a standard deletion-AOPC curve under two removal operators. Under zero-masking the four methods order IG (0.4776) > Saliency (0.4730) > DeepSHAP (0.4724) > Occlusion (0.4629), a spread of 0.0147; under PacketDO they collapse to within 0.0010, below the per-seed resampling noise (~0.0058). The Kendall tau between the two orderings is 0.333. The robust and load-bearing effect is not the full four-way reordering (the middle ranks differ by less than one test-sample count) but a specific demotion: under zero-masking, occlusion — the method with zero false-confidence and the best necessity-rank correlation (0.715) — is ranked last, below saliency, whose false-confidence is 0.667 and whose necessity correlation is 0.043. The protocol- invalid operator manufactures an ordering that inverts the audit evidence, and it does so on non-monotone deletion curves computed on impossible packets. Any comparison of traffic-classifier explainers that used the zero-masking operator — the field default — therefore inherits an ordering that is an artifact of the operator, not a property of the explanations (contribution C4).

Table III. E5: deletion AOPC of each explainer under zero-mask vs PacketDO removal (p=1.0 ByteCNN; PacketDO mean over 5 resampling seeds). Higher AOPC = steeper deletion = nominally more faithful.

explainer	zero-mask AOPC	PacketDO AOPC	false-conf (seed 0)
IntegratedGradients	0.4776	0.4882	0.0
Saliency	0.4730	0.4875	0.667
DeepSHAP	0.4724	0.4872	0.0
Occlusion	0.4629	0.4881	0.0

Under zero-mask, Occlusion (false-confidence 0) is ranked last; under PacketDO the four are within 0.001 (below resampling noise ~0.006).

7. Discussion and implications

The failure is added falsehood, not missed truth. Across every model, dataset, and planting strength, the explainers we tested rank a model's genuinely-used feature at or near the top: on the synthetic benchmark precision-at-k is 1.0, and on real CIC-IDS2017 the destination-port shortcut that the model most depends on is surfaced by per-class SHAP. What the explainers add is confident, high-magnitude importance for features the model provably does not use. On the synthetic benchmark this shows up as gradient-saliency false-confidence around 0.7; on real data it is starker, because the redundant features are numerous: nine of TreeSHAP's global top ten are packet-length statistics that are collectively load-bearing but individually near-zero in interventional necessity, and they outrank the single most necessary feature. An operator reading the explanation cannot separate the feature the model uses from the many it merely could have used. This is a more dangerous failure than a low-quality but honest ranking, because the false positives are exactly the plausible-looking features that invite a wrong intervention.

Redundancy makes single-explanation evaluation ill-posed, and exactness does not help. The reason the false positives appear is structural: traffic data encodes the same discriminative signal many times over — two headers that both leak the class, a family of correlated flow statistics — so a model is free to commit to one encoding and ignore the rest, and different models (or the same architecture under a different seed) commit differently. An attribution method that returns a single importance vector is answering a question with no unique answer. TreeSHAP makes this precise: it computes exact Shapley values, and it still fabricates confidence in an unused shortcut, because Shapley values distribute credit according to the correlation structure of the data rather than the realized reliance of the model. Exactness of the attribution is orthogonal to its faithfulness when the explanandum is not unique. Our redundancy degree R(M) is the property that has to be measured before a single-vector explanation can even be interpreted, and it is measurable only by intervention.

The removal operator is not a detail. E5 shows that the ranking of explainers by a standard deletion-style faithfulness score can invert depending on whether features are removed by zero-masking or by PacketDO: under the protocol-invalid zero-mask operator the method with zero false-confidence (occlusion) is ranked worst, below a method with high false-confidence, because the zero-masked inputs are off-manifold and the resulting deletion curves are not even monotone. Under the protocol-valid operator the methods that all correctly identify the model's single true shortcut collapse, correctly, to a spread below the resampling noise.

Any comparison of explainers for traffic classifiers that used the zero-masking operator — the field default — therefore inherits an ordering that is an artifact of the operator, not a property of the explanations. The same off-manifold mechanism reappears inside an explainer: LIME's false-confidence traces to its perturbation kernel sampling constant protocol fields with unit variance (an artifact of standardizing zero-variance features), i.e. LIME evaluates the model on impossible packets as part of its own definition.

Why this matters operationally. The traffic-classification literature does not stop at displaying attributions. Deployed and published systems act on them: xNIDS [18] generates active intrusion-response rules from its explanations, ShortcutCatcher [15] deletes features from the training pipeline based on what an explainer flags, and analyst-facing tools present attributed features as the justification for an alert. A false-confidence feature in that setting is not a mislabeled figure; it is a firewall rule keyed on a coincidence, a genuinely-predictive feature wrongly pruned, or an analyst's trust spent on the wrong evidence. The interventional reference we provide is the check these pipelines currently lack: before an attribution is allowed to drive an action, it can be scored against what the model actually depends on.

What we do not claim. We do not claim explanations are worthless, nor that any single method is uniformly best. Intervention-aligned methods (occlusion, and DeepSHAP at sufficient signal strength) are the most faithful in our study, but occlusion's advantage is partly structural — it removes features much as our ground-truth operator does, so its clean record is not independent evidence (Section 8) — and even it is only as good as the removal operator it uses; DeepSHAP fails at weak planting strength in a subset of seeds. The constructive reading of our results is that faithfulness for traffic classifiers is achievable, but only with a protocol-valid intervention and an explicit accounting of redundancy; neither is present in current practice.

8. Threats to validity

Resampling changes the joint distribution. PacketDO preserves each field's marginal but, by sampling it independently of the label, alters its joint distribution with the other fields. A model that relied on a genuine correlation between two fields would register a necessity drop that is real but not attributable to either field alone. We mitigate this in three ways: interventions resample class-agnostically from real values (so every counterfactual is a value the field actually takes); we report field-set necessity for the documented redundant families, not only single-field necessity; and we include a null-intervention control field in every table, whose necessity stays within the noise floor and calibrates what "zero necessity" means for the estimator.

The model must not be retrained under intervention. Necessity and sufficiency are properties of a fixed model; retraining after an intervention (as remove-and-retrain protocols do [8]) measures a different model and reintroduces the confound our operator was designed to avoid. We freeze the model for all interventions. Where a strength sweep requires different models, each strength is trained once before any intervention is applied to it.

Byte-versus-field granularity. Byte-level explainers attribute to byte offsets, which we aggregate to protocol fields using a fixed header-offset map valid for the canonical no-options layout. Two consequences: attributions to bytes shared by two fields, or to variable-length options, are aggregated approximately; and where header lengths differ across samples (the misalignment that motivates protocol-aware parsing) absolute offsets are not stable. We report results at both byte and field granularity and use protocol-parsed offsets for the misaligned cases.

Cost of exact methods. KernelSHAP is orders of magnitude slower than the gradient and tree methods; we bound it with stratified subsampling and report the measured wall-clock cost, which is itself a datapoint for line-rate deployment. This means KernelSHAP results are computed on fewer samples than the other methods, with correspondingly wider uncertainty.

Occlusion is close to the ground-truth operator. Occlusion removes a feature by permutation, structurally similar to PacketDO's resampling. Its strong faithfulness is therefore partly by construction, and we state this explicitly rather than presenting occlusion as an independent winner. E5 disentangles the effect by scoring occlusion under both the zero-mask and PacketDO operators.

Synthetic versus real. The graded-strength results are on synthetic traffic, where ground truth is planted and exact. Their external validity rests on the real-data replication (CIC-IDS2017), where the false-confidence and blind-spot phenomena reproduce on documented natural artifacts; but the real data is flow-level, so the byte-level artifacts (time-to-live, header misalignment, sequence-number bytes) that require packet captures are documented and left to a byte-level replication. The destination-port and flow-construction artifacts we do measure on real data are the ones representable in flow features.

Table IV. A2 real-data (CIC-IDS2017) destination-port robustness: 4 seeds x default RF (d20/l20) + 2 alt-config runs (d30/l5).

run	N(dst) acc-rank /52	TreeSHAP rank /52	acc-only FC /10	Brute-Force recall drop
default_seed0	1	5	9	0.378
default_seed1	1	7	8	0.441
default_seed2	1	8	9	0.382
default_seed3	1	9	9	0.199
alt_d30l5_seed0	1	5	9	0.360
alt_d30l5_seed1	1	6	9	0.356

Necessity rank 1/52 in all six runs; TreeSHAP never ranks it better than 5th.

Serializer circularity. Both PacketDO's validity and the validity checker rely on the same protocol serializer. We harden this in two ways: the checker was cross-validated against a from-scratch one's-complement (RFC 1071) checksum implementation on adversarial corruptions, and the operator's per-field correctness is covered by unit tests; an independent validator (e.g. tshark) is a further step. The E1 macro-validity figure is an unweighted average over synthetic fields, not a per-packet real-traffic rate, and three fields are valid under zero-masking only because the generator left them zero; on real traffic the zero-mask validity would if anything be lower, so the reported gap is conservative.

Seed sensitivity of quantized metrics. False-confidence is a fraction over a small set of named fields and therefore takes quantized values (0, 1/3, 1/2, …); a small rescaling of attributions can move it between levels. We report it as a mean with standard deviation over five seeds and give the per-seed values, and we distinguish claims that hold in every seed (saliency false-confidence, occlusion's clean record) from those that are seed-contingent (the exact strength at which Integrated Gradients or DeepSHAP first fabricates importance). On the real-data side, the six-run sweep spans four seeds and two forest configurations; the alternative (deeper) configuration strengthens rather than weakens the findings, so the artifact reliance is not an under-fitting effect of the default configuration.

9. Limitations and future work

Our scope is the faithfulness axis of explanation quality for traffic classifiers, evaluated with a protocol-valid intervention. Several adjacent problems, identified in our systematic gap analysis of the literature (see the supplementary gap-analysis document accompanying the artifact release), are deliberately out of scope and define the research programme this instrument enables:

- **Metric meta-evaluation (diagnosticity).** With planted-truth models in hand, the field's existing faithfulness *metrics* — deletion-AUC, descriptive accuracy, fidelity — can themselves be scored on whether they distinguish faithful from unfaithful explainers. E5 is a first datapoint (the zero-mask AOPC ranking inverts the audit evidence); a systematic diagnosticity study is the natural sequel to this paper.
- **Drift-aware evaluation.** Traffic distributions drift, and an explanation that is faithful at training time may not remain so under deployment drift [34]. No temporal-drift-aware explanation evaluation exists for traffic; our interventional reference is recomputable at any time slice and could anchor one.
- **The plausibility axis and analyst studies.** We measure whether explanations reflect the model, not whether they help an analyst [5], [21]. A two-axis protocol that scores plausibility and faithfulness separately — and the analyst-grounded human evaluation the security setting ultimately requires — are open, and our ground-truth tables provide the faithfulness half of such a protocol.
- **Adversarial robustness of explainers.** An adversary can manipulate explanations [35]; whether protocol-valid perturbations can be crafted to mislead traffic-model explainers, and whether interventional auditing detects such manipulation, is unstudied.
- **Full redundancy (Rashomon) enumeration.** Our R(M) counts disjoint sufficient field sets by greedy interventional removal; a complete enumeration of the Rashomon set of equally-performing models and their explanation multiplicity is a larger undertaking that our credit-splitting results motivate.
- **Attention as explanation, and deep sequence models.** The audit covers gradient, SHAP-family, perturbation, and tree methods on a byte-CNN and a random forest. Extending it to attention-based transformer classifiers (e.g. ET-BERT [3]) — where attention-as-explanation remains contested — and to a byte-level replication on packet-capture datasets with documented misalignment artifacts (ISCX VPN-nonVPN [36]) is future work; both are mechanical extensions of the released benchmark.

10. Reproducibility

Every number in this paper traces to a versioned result file produced by a seeded script, and the full pipeline runs on commodity hardware (CPU for all tree-model and intervention experiments; a consumer laptop GPU suffices for the ByteCNN and the perturbation explainers). The artifact release contains: the PacketDO operator with its per-field unit-test suite (42 passed, 1 skipped) and the RFC 1071 cross-validated validity checker; the synthetic generator and per-strength planted-shortcut datasets (seeded); the benchmark driver that trains both model families, computes interventional ground truth, runs all eight explainers, and audits them; the E5 operator-sensitivity driver; and the CIC-IDS2017 real-data pipeline with its six-run robustness harness (approximately 200–220 s per run on CPU; 1,259 s for all six). Multi-seed aggregation is performed by a script that validates the provenance of every per-seed result file (internal seed and strength fields must match the filename; mismatched files are excluded and the run exits nonzero) — a guard added after it caught a file-duplication fault in an early automated run — and callers are required to check its exit code. The seed-0 real-data reference run is reproduced float-identically by the robustness harness's first run. All datasets are public (CIC-IDS2017 original and Engelen-corrected [11]); the synthetic traffic is regenerated from a seed. One top-level make target reproduces every table and figure from a clean checkout.

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